



BHAVAN'S PUBLIC SCHOOL, DOHA-QATAR

MIDTERM EXAMINATION MODEL (2026-'27)

ANSWERKEY MATHEMATICS

CLASS: VI

**Time: 3 hours
Maximum Marks: 80**

General Instructions:

1. This Question Paper has 4 Sections A, B, C and D.
2. Section A has 19 MCQs and 1 assertion-reason based question of 1 mark each.
3. Section B has 6 questions carrying 02 marks each.
4. Section C has 8 questions carrying 03 marks each.
5. Section D has 6 questions carrying 04 marks each.

SECTION A

(Multiple Choice Questions- Each question carries 1 mark)

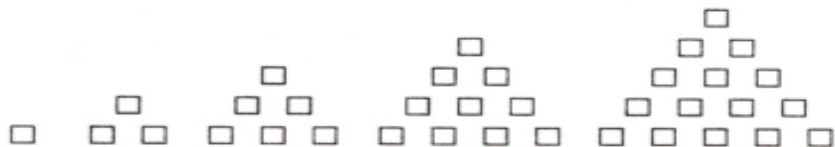
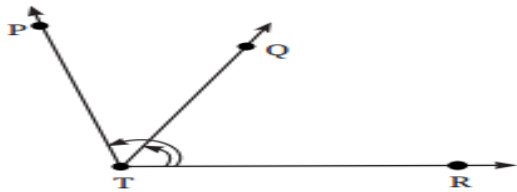
Qn No		Marks
1	b. 6	1
2	c. Reflex	1
3	c)Line segment	1
4	b) bar graph	1
5	c) angle	1
6	b) horizontal	1
7	b) 205	1
8	d)100001	1
9	b) 5	1
10	b) subtraction	1

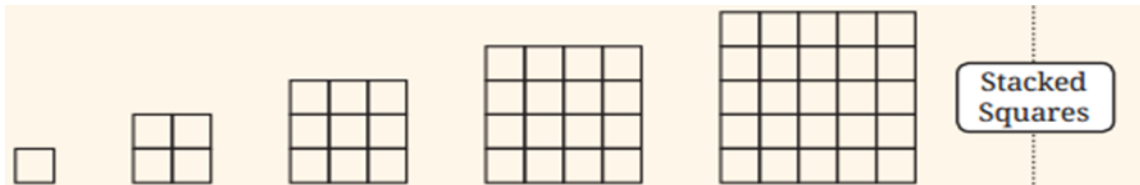
11	c)co-prime numbers	1
12	b) 125	1
13	b) 15	1
14	b) protractor	1
15	c) 871	1
16	b) 20	1
17	d) 360°	1
18	a)2	1
19	c)4	1
20	a) Both A and R are true, and R is the correct explanation of A..	1

SECTION – B

This section comprises of very short answer type questions of 2 marks each

21	a) 546840 Ans)546840 → Last three digits are 840. Since $840 \div 8 = 105$, it is divisible by 8. So, 546840 is divisible by 8. b) 786111 Ans) 786111 → Last three digits are 111. Since 111 is not divisible by 8. So 786111 is not divisible by 8.	2
22	<u>Adding counting numbers up</u> $1 = 1$ $1 + 2 = 3$	2

	$1 + 2 + 3 = 6$ $1 + 2 + 3 + 4 = 10$ $1 + 2 + 3 + 4 + 5 = 15$ and so on. Here, we get sequence of triangular numbers. A smaller pictorial explanation is 	
23	Factors of 20: 1, 2, 4, 5, 10, 20 Factors of 28: 1, 2, 4, 7, 14, 28 Therefore, the common factors are 1, 2, and 4.	2
24	 In the given figure, marked angles are $\angle RTQ$ and $\angle RTP$	2

25	<table><tr><td>2</td><td>456</td></tr><tr><td>2</td><td>228</td></tr><tr><td>2</td><td>114</td></tr><tr><td>3</td><td>57</td></tr><tr><td>19</td><td>19</td></tr><tr><td></td><td>1</td></tr></table> $456 = 2 \times 2 \times 2 \times 3 \times 19$	2	456	2	228	2	114	3	57	19	19		1	2
2	456													
2	228													
2	114													
3	57													
19	19													
	1													
26	a) The number of squares in each shape of the sequence of stacked squares is: 1, 4, 9, 16,.....  b) The number sequence given by this is that of perfect square numbers.	2												

SECTION - C

This section comprises of short answer type questions of 3 marks each

27

a)

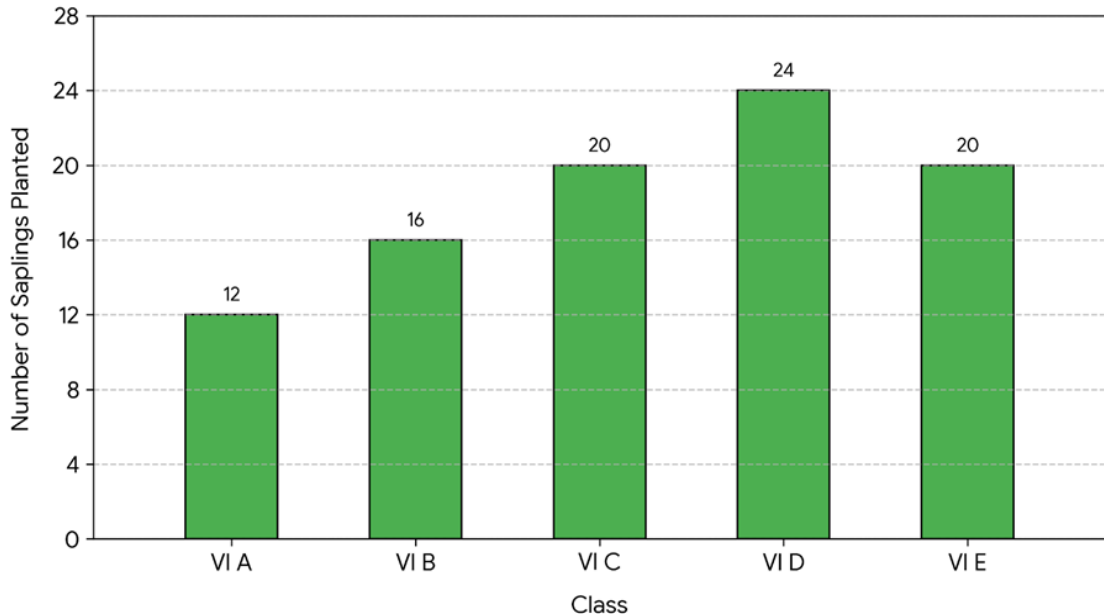
3

Diagram showing a sequence of numbers in a row: 987, 457, 999, 123, 888, 345, 789. Red arrows point from the blue-highlighted numbers (987, 999, 888, 789) to boxes labeled "supercell".

b)

Diagram showing four hexagonal numbers represented by dots. The first has 1 dot (labeled 1). The second has 7 dots (labeled 7). The third has 19 dots (labeled 19). The fourth has 37 dots (labeled 37).

Ans) 1, 7, 19, 37, are hexagonal numbers

28	Environment Day Activity: Saplings Planted by Class (Scale: 1 unit length = 4 saplings)  <table><caption>Data for Environment Day Activity: Saplings Planted by Class</caption><tr><th>Class</th><th>Number of Saplings Planted</th></tr><tr><td>VI A</td><td>12</td></tr><tr><td>VI B</td><td>16</td></tr><tr><td>VI C</td><td>20</td></tr><tr><td>VI D</td><td>24</td></tr><tr><td>VI E</td><td>20</td></tr></table>	Class	Number of Saplings Planted	VI A	12	VI B	16	VI C	20	VI D	24	VI E	20	3
Class	Number of Saplings Planted													
VI A	12													
VI B	16													
VI C	20													
VI D	24													
VI E	20													
29	The smallest 5-digit palindrome is 10001. The largest 5-digit palindrome is 99999. Sum: $10001 + 99999 = 110000$ Difference: $99999 - 10001 = 89998$	3												
30	a) Mathematics Club= 5 symbols $\times$ 5 = 25 students b) Sports Club. c) $30-15 =$ 15 students d) Total students = $18 \times 5 = 90$ students.	3												
31	a): 14 and 25 • Prime factorization of 14 = 2×7 • Prime factorization of 25 = 5×5 • Common prime factors: None • Common factor =1 • So 14 and 25 are co-prime numbers b): 22 and 44 • Prime factorization of 22 = 2×11	3												

- Prime factorization of $44 = 2 \times 2 \times 11$
- **Common prime factors:** 2 and 11
- So 22 and 44 are **not co-prime numbers**.

c): 28 and 45

- Prime factorization of $28 = 2 \times 2 \times 7$
- Prime factorization of $45 = 3 \times 3 \times 5$
- **Common prime factors:** None
- Common factor = 1
- So, 28 and 45 are co-prime numbers.

32

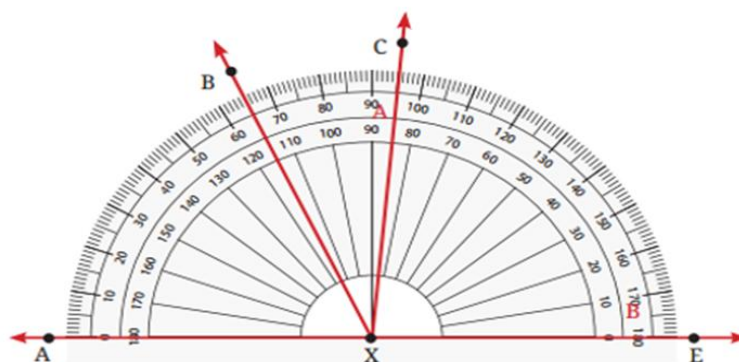
 = 2 books

Student	Number of story books
Anu	
Diya	
Riya	
Kabir	
Meera	

3

33

$\angle BXE = 115^\circ$
 $\angle CXE = 85^\circ$
 $\angle AXB = 65^\circ$
 $\angle BXC = 30^\circ$



3

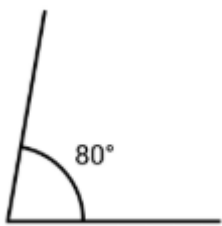
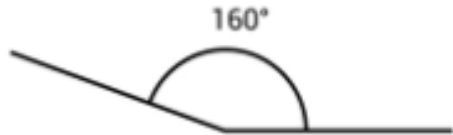
34		1																					
	<table border="1"> <thead> <tr> <th>MARKS</th><th>TALLY MARKS</th><th>FREQUENCY</th></tr> </thead> <tbody> <tr> <td>5</td><td></td><td>6</td></tr> <tr> <td>6</td><td></td><td>2</td></tr> <tr> <td>7</td><td></td><td>4</td></tr> <tr> <td>8</td><td></td><td>5</td></tr> <tr> <td>9</td><td></td><td>1</td></tr> <tr> <td>10</td><td></td><td>2</td></tr> </tbody> </table>	MARKS	TALLY MARKS	FREQUENCY	5		6	6		2	7		4	8		5	9		1	10		2	
MARKS	TALLY MARKS	FREQUENCY																					
5		6																					
6		2																					
7		4																					
8		5																					
9		1																					
10		2																					

SECTION D

This section comprises of long answer- type questions of 4 marks each

35	Using the Collatz rule: - If the number is even, divide it by 2. - If the number is odd, multiply it by 3 and add 1. Starting with 21: $21 \times 3 + 1 = 64$ $64 \div 2 = 32$ $32 \div 2 = 16$ $16 \div 2 = 8$ $8 \div 2 = 4$ $4 \div 2 = 2$ $2 \div 2 = 1$ Therefore, the sequence reaches 1, so the Collatz Conjecture holds for the starting number 21. So, the sequence is $21 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$	4
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36	a) Prime factorisation of 72 $72 = 2 \times 2 \times 2 \times 3 \times 3$ Prime factorisation of 18 $18 = 2 \times 3 \times 3$ Since all the prime factors of 18 are contained in the prime factorisation of 72, $72 \div 18 = 4$ Therefore, 72 is divisible by 18. b) i) 180° – Straight angle ii) 50° – Acute angle iii) 90° – Right angle iv) 130° – Obtuse angle	4
37	a) Using the Kaprekar routine: $3524 \rightarrow 5432 - 2345 = 3087$ $3087 \rightarrow 8730 - 0378 = 8352$ $8352 \rightarrow 8532 - 2358 = 6174$ So, 3524 takes 3 rounds to reach the Kaprekar constant 6174 b) No, 3366 is not divisible by 4. Because ,For divisibility by 4, we check the last two digits. The last two digits are 66, and 66 is not divisible by 4 So ,3366 is not divisible by 4.	4

38	a) Use a protractor to draw angles having the following degree measures: i) 80°  ii) 160°  b) The composite numbers from 70 to 80 are: 70, 72, 74, 75, 76, 77, 78, 80.	4
39	<u>Case Study Question</u> Answers a) 36 flowers b) Square number pattern c) 16 d) 36 and 49	4
40	<u>Case Study Question</u> Answers: a) 17, 29 b) 12, 21, 35, 42 c) 1, 3, 7, 21 d) Prime number	4

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